

Introduction

A Mark Sheet Management System is a simple application used by educational institutions to store and manage student academic results. It records student names, subjects, marks, and computes totals, averages, and grades. This program aims to reduce manual work, ensure accuracy, and enable quick access to result data.

☀ Features of the Application

Add student details

Enter subject-wise marks

Calculate total, average, and grade

Display the mark sheet

Error handling for invalid inputs

🧠 Java Concepts Used

Java Concept Usage

Classes and Objects Student and MarkSheet classes encapsulate data and behaviors.

Constructors Initialize student data and marks.

Arrays Store multiple subject marks.

Loops For processing multiple subjects and records.

If-Else Grade calculation logic.

Methods To organize code for computation and display.

OOP principles Encapsulation and modularity applied using separate classes.

Java Code (Console-Based Mark Sheet)

```
import java.util.Scanner;

public class Marksheet {

    // User-defined method to calculate total
    public static int calculateTotal(int[] marks) {
        int total = 0;
        for (int mark : marks) {
            total += mark;
        }
        return total;
    }

    // User-defined method to calculate grade
    public static String calculateGrade(double percentage) {
        if (percentage >= 90) return "A+";
        else if (percentage >= 80) return "A";
        else if (percentage >= 70) return "B";
        else if (percentage >= 60) return "C";
        else if (percentage >= 50) return "D";
        else return "F";
    }

    // User-defined method to display marksheet
    public static void displayMarksheet(String name, int roll, int[] marks, int total, double
percentage, String grade) {
        System.out.println("\n===== MARKSHEET =====");
        System.out.println("Name      : " + name);
        System.out.println("Roll No   : " + roll);
        System.out.println("-----");
        for (int i = 0; i < marks.length; i++) {
            System.out.println("Subject " + (i + 1) + " : " + marks[i]);
        }
        System.out.println("-----");
        System.out.println("Total Marks : " + total);
        System.out.println("Percentage : " + percentage + "%");
        System.out.println("Grade      : " + grade);
        System.out.println("=====\\n");
    }

    public static void main(String[] args) {
```

```

Scanner scanner = new Scanner(System.in);

// Input student details
System.out.print("Enter Student Name: ");
String name = scanner.nextLine();

System.out.print("Enter Roll Number: ");
int roll = scanner.nextInt();

int[] marks = new int[5]; // 5 subjects
int i = 0;

// Input marks using while loop
while (i < 5) {
    System.out.print("Enter marks for Subject " + (i + 1) + ": ");
    marks[i] = scanner.nextInt();

    if (marks[i] < 0 || marks[i] > 100) {
        System.out.println("Invalid mark! Must be between 0 and 100.");
        continue; // retry this subject
    }
    i++;
}

// Calculations
int total = calculateTotal(marks);
double percentage = total / 5.0;
String grade = calculateGrade(percentage);

// Display
displayMarksheet(name, roll, marks, total, percentage, grade);
}
}

```

